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(54) **A METHOD OF INSTALLING A CABLE  
BOLT, A SYSTEM AND COMPONENTS OF  
SAME**

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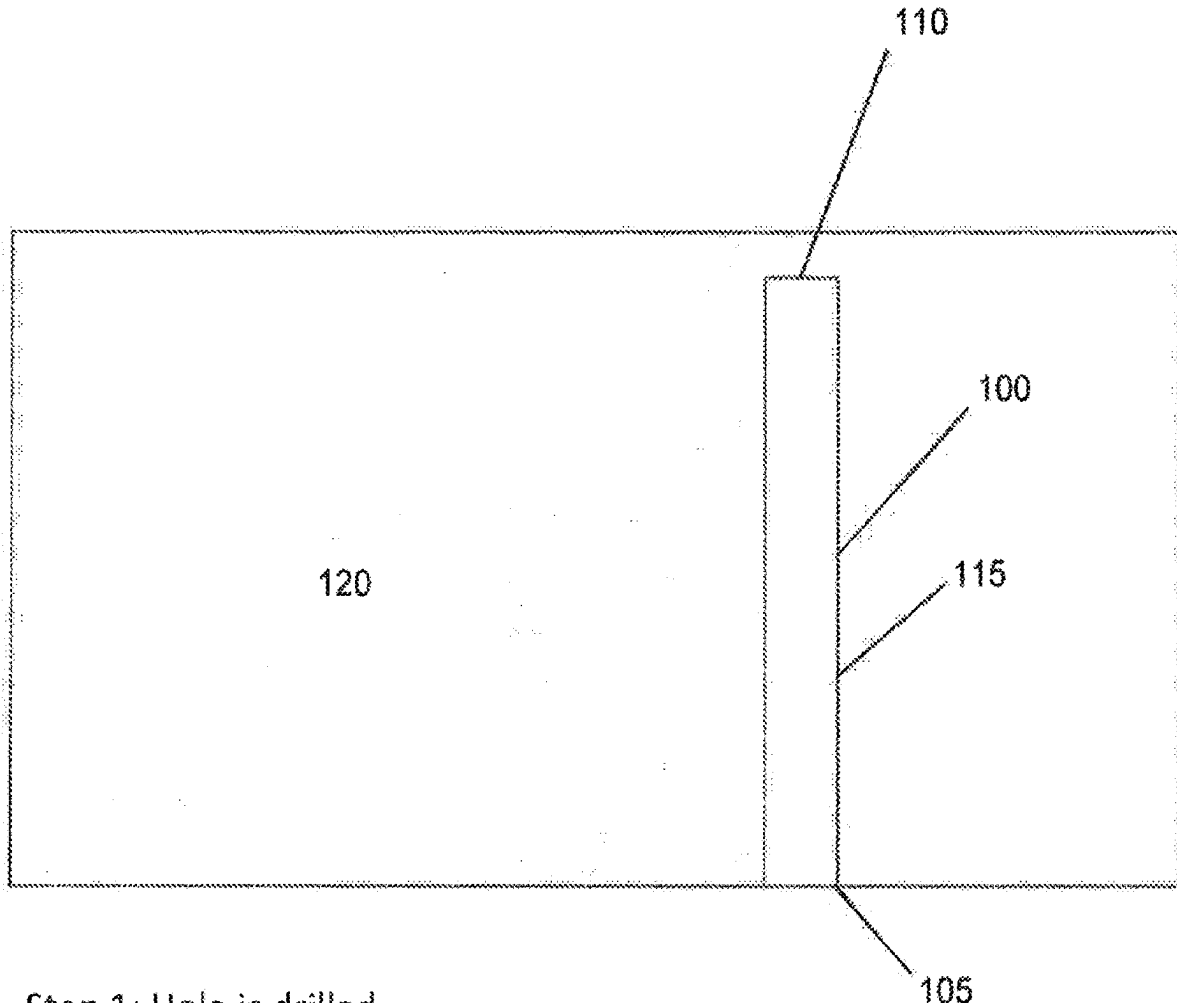
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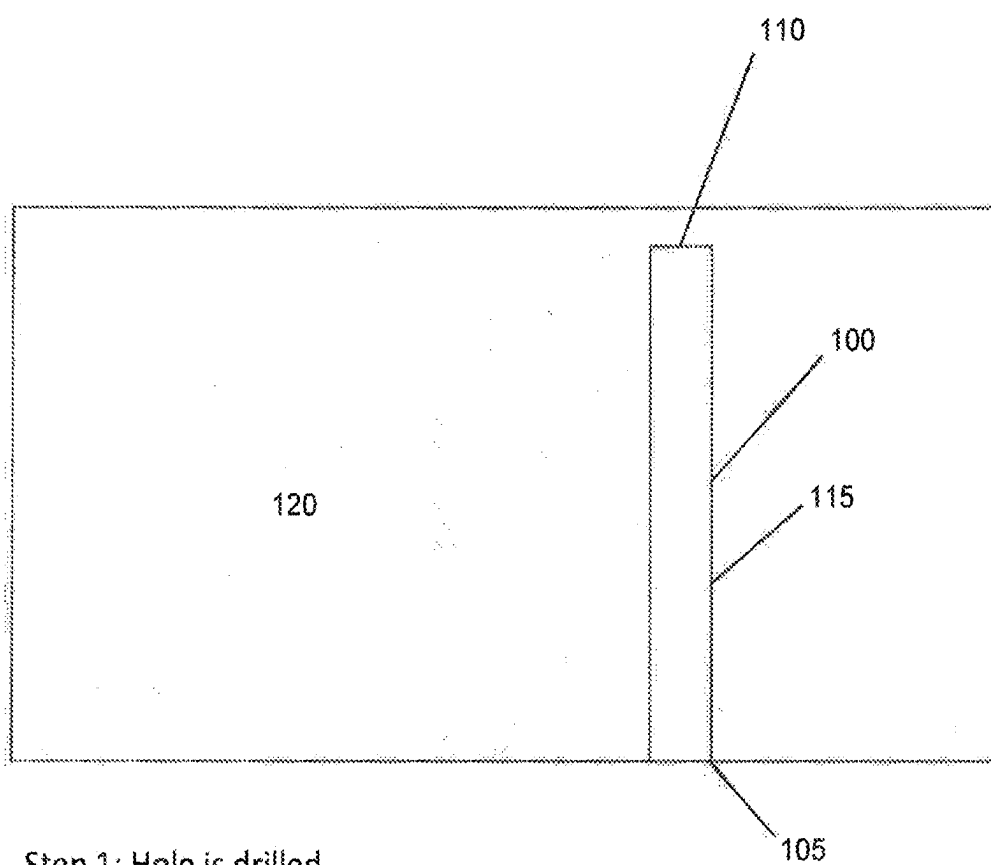
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**ABSTRACT**

A method of installing a grouted cable bolt, a system for providing same and components which may form part of the method and/or system including a delivery assembly comprising a delivery nozzle at a distal end of the delivery assembly; a mixing chamber in fluid communication with the delivery nozzle; a first line for conveying a first component to the mixing chamber; and, a second line for conveying a second component to the mixing chamber the first line being located within the second line, and wherein the hardenable composition in liquid form is formed from mixing the first component with the second component.



**Step 1: Hole is drilled  
in the rock**



Step 1: Hole is drilled  
in the rock

Figure 1

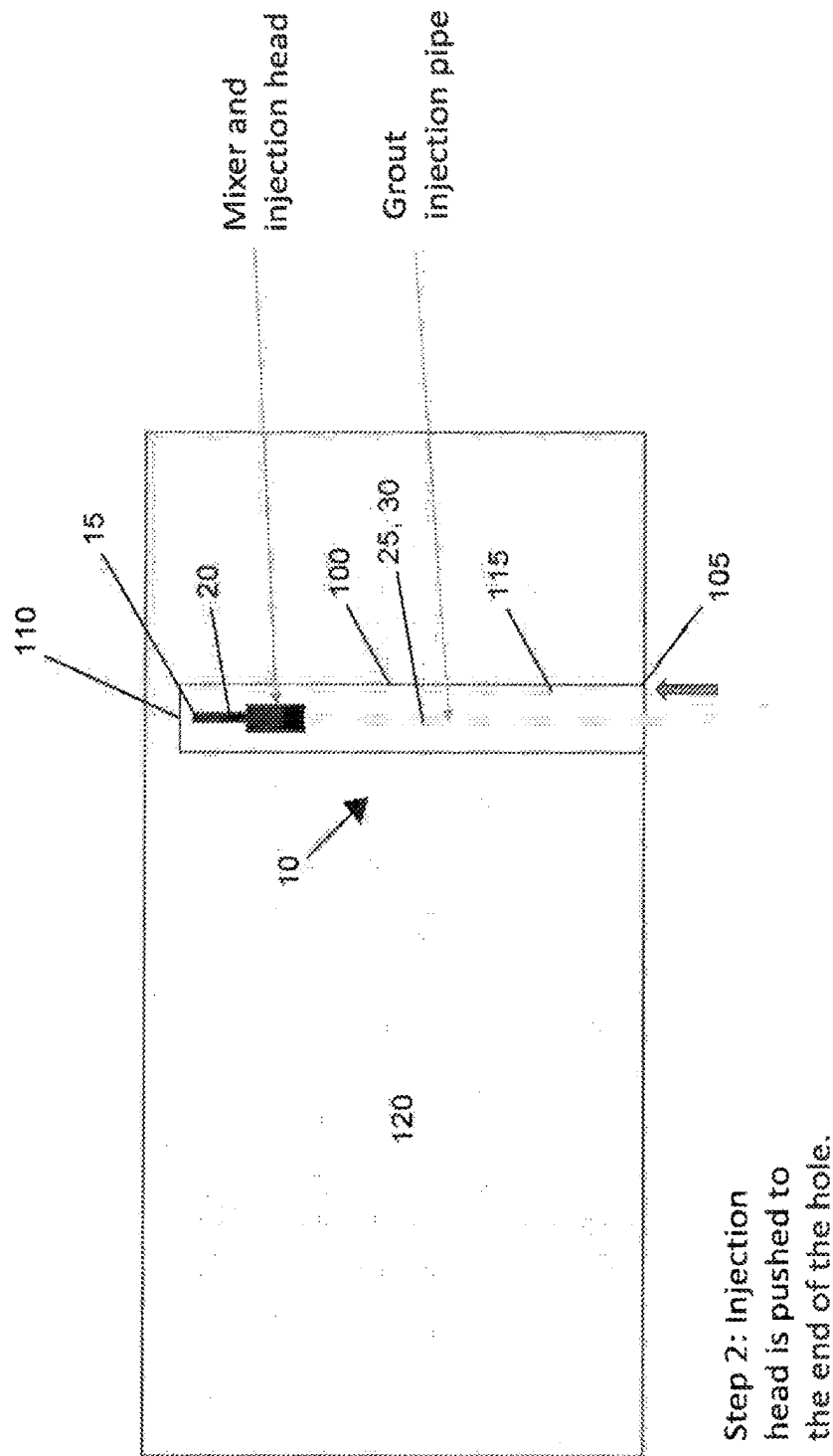


Figure 2

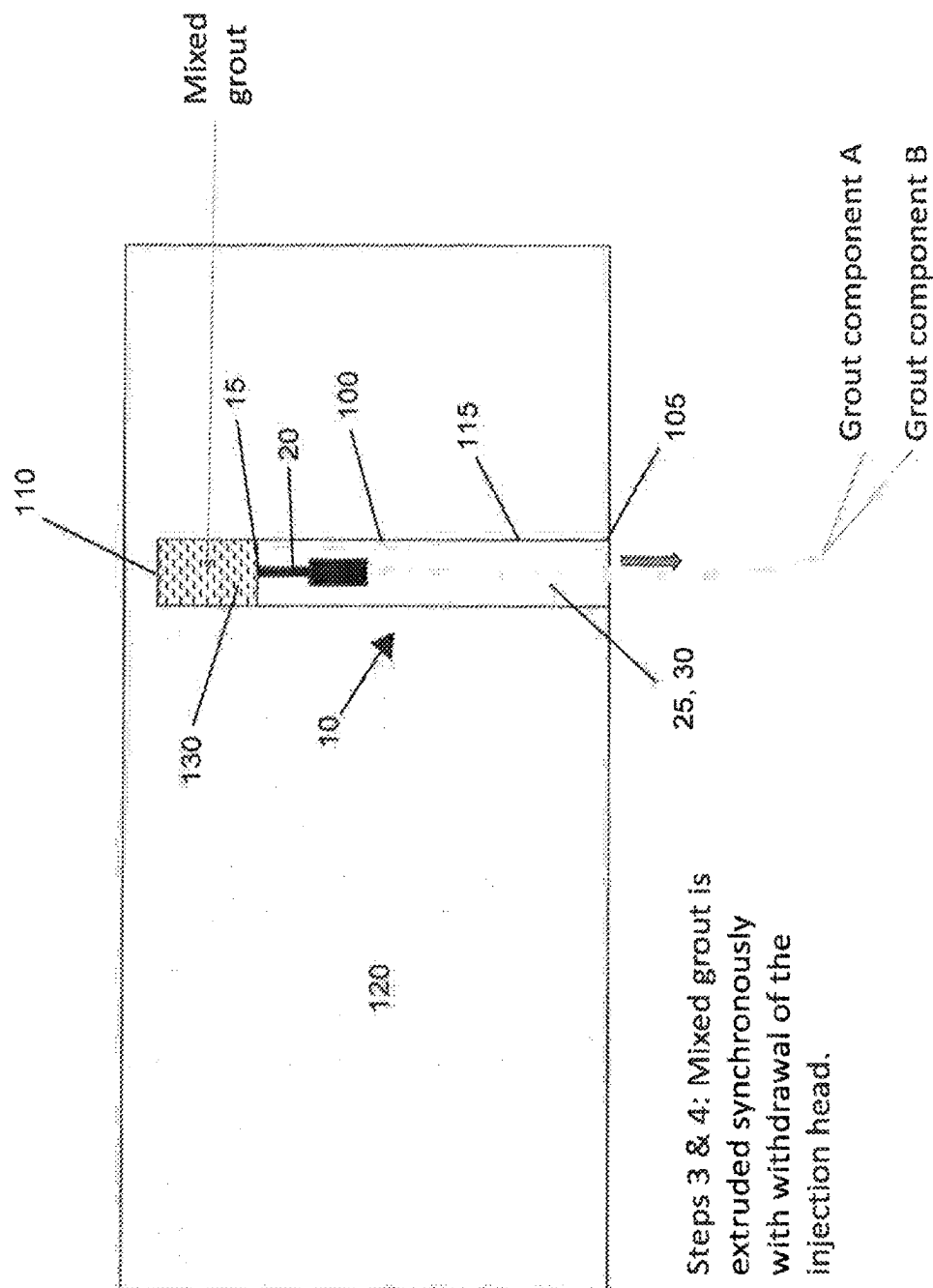


Figure 3

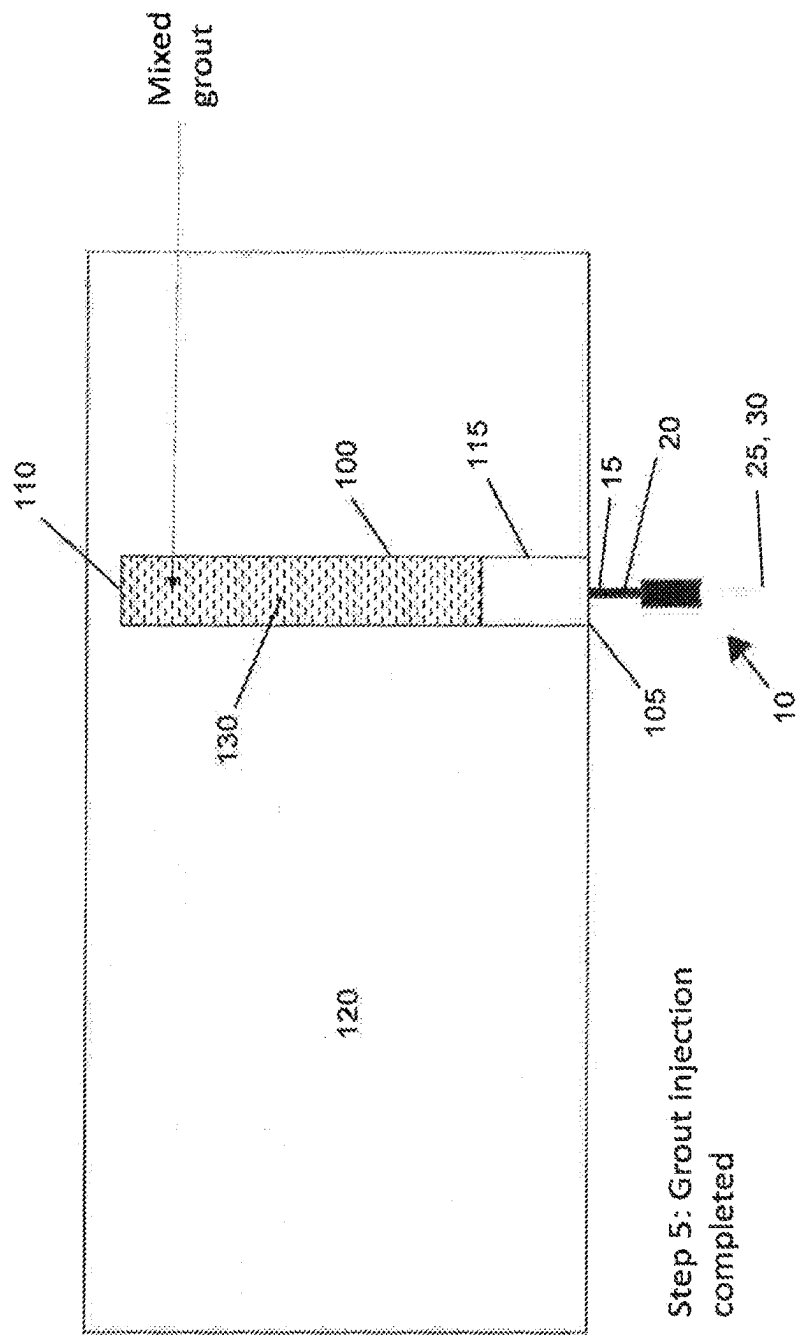


Figure 4

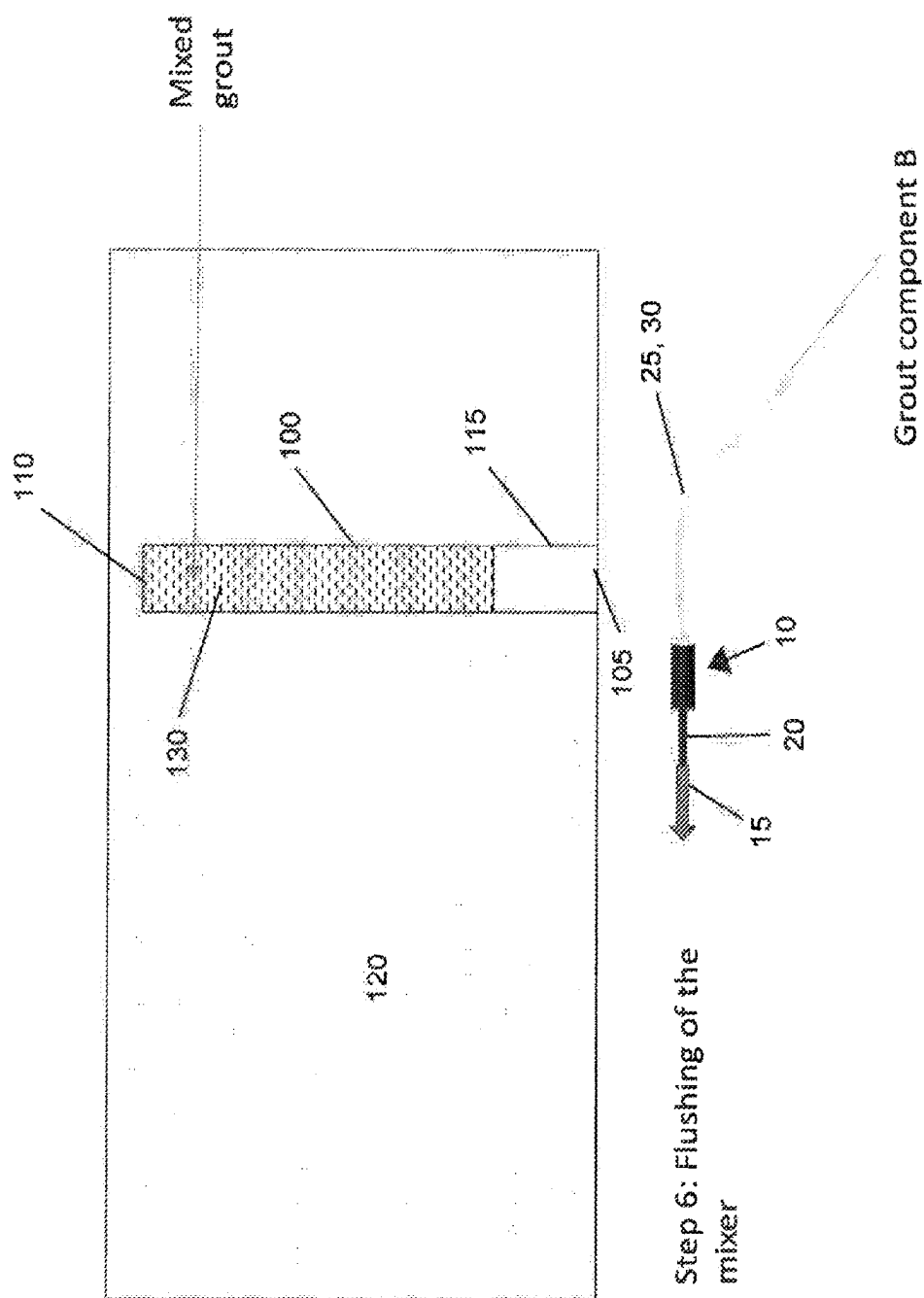


Figure 5

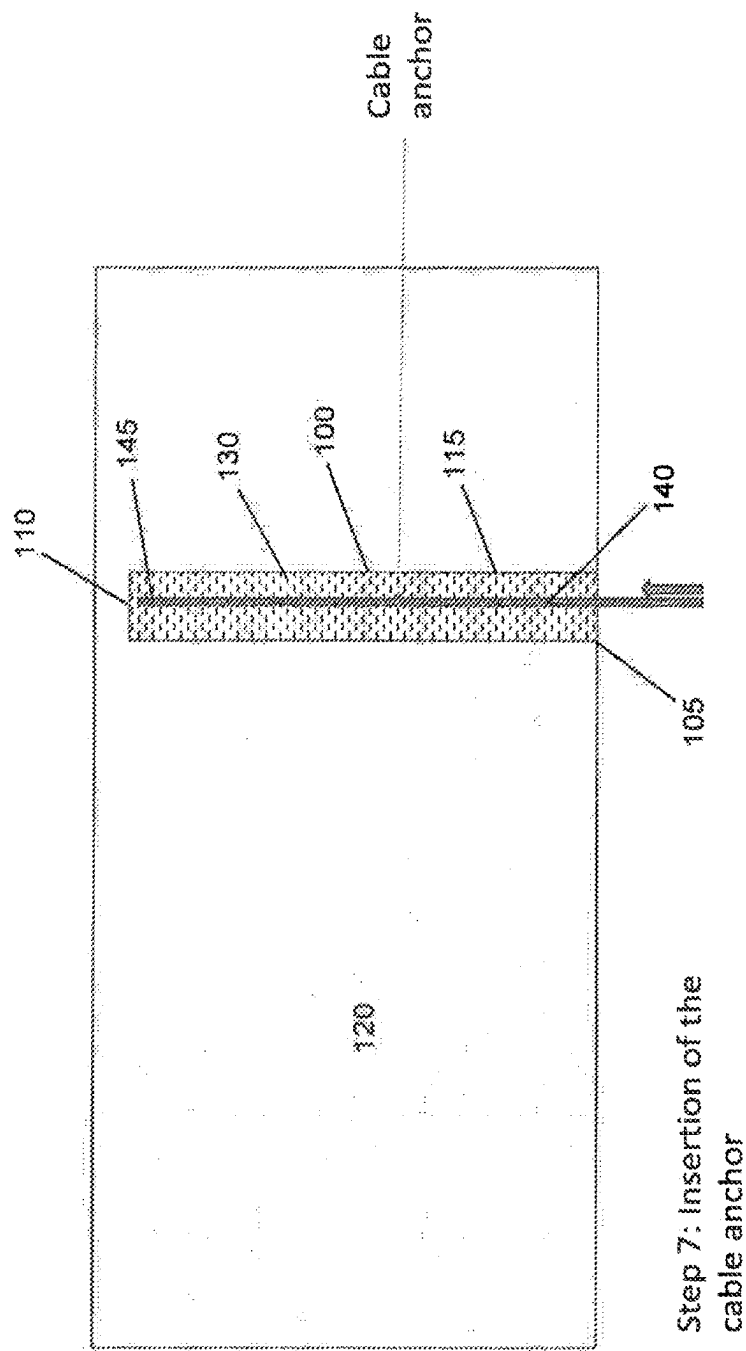


Figure 6

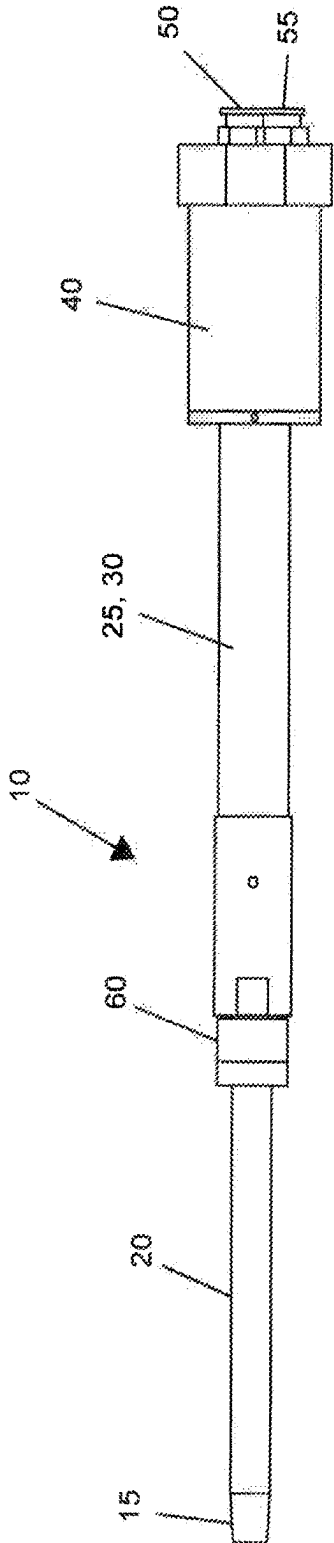


Figure 7

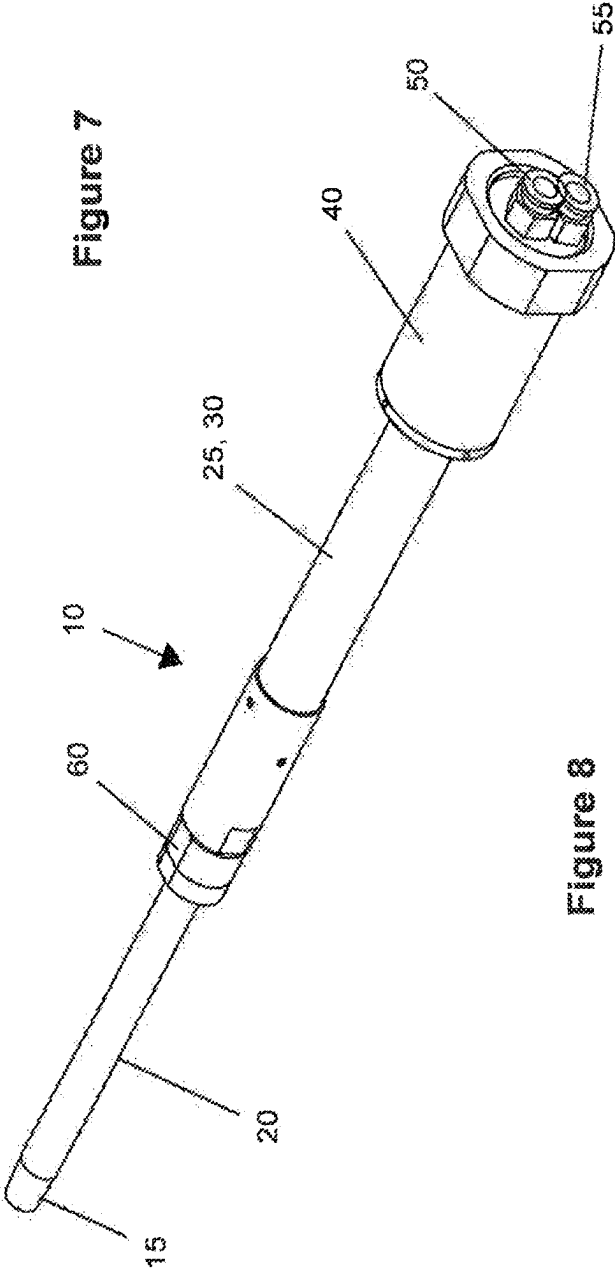


Figure 8



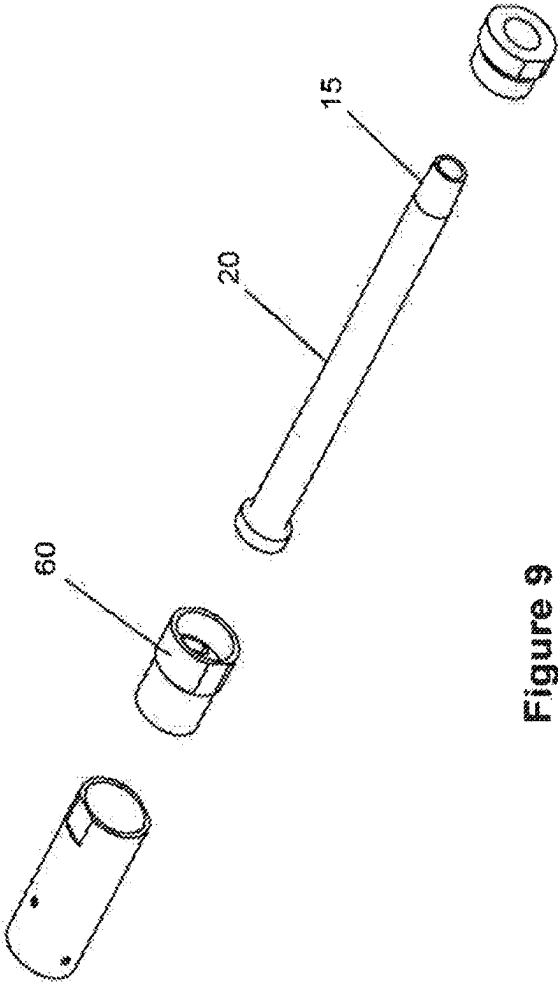


Figure 9

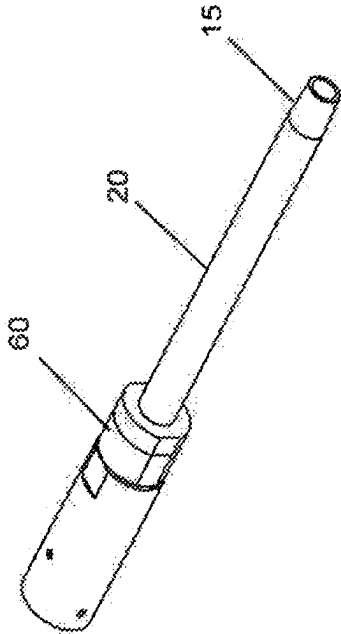


Figure 10

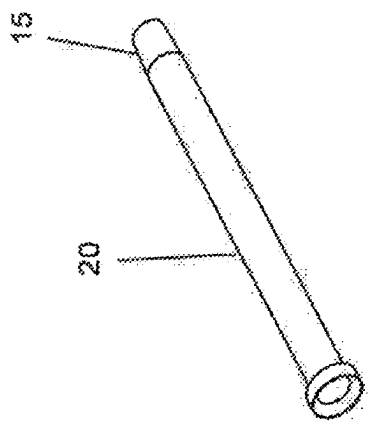


Figure 11

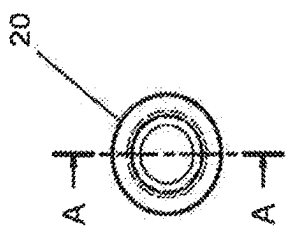


Figure 12

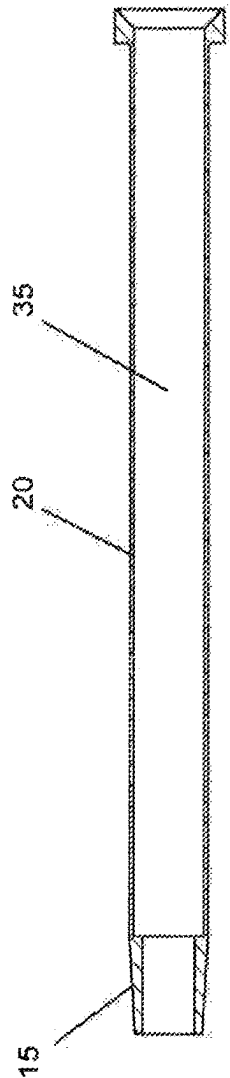


Figure 13

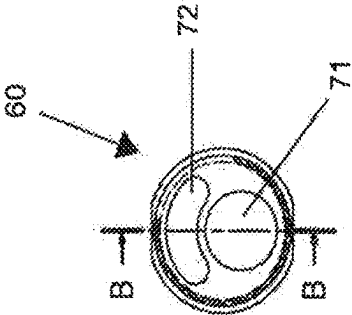


Figure 15

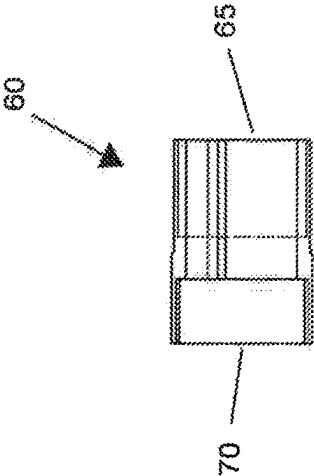


Figure 16

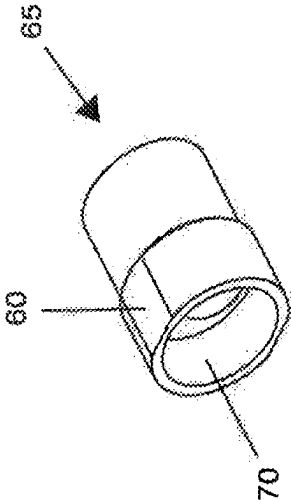


Figure 14

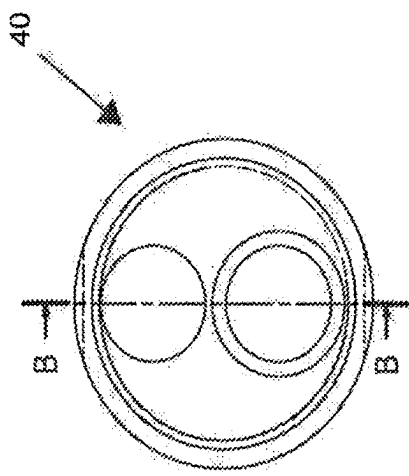


Figure 17

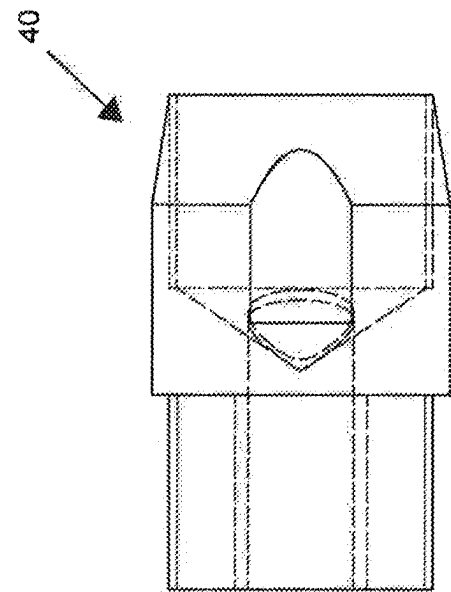


Figure 18

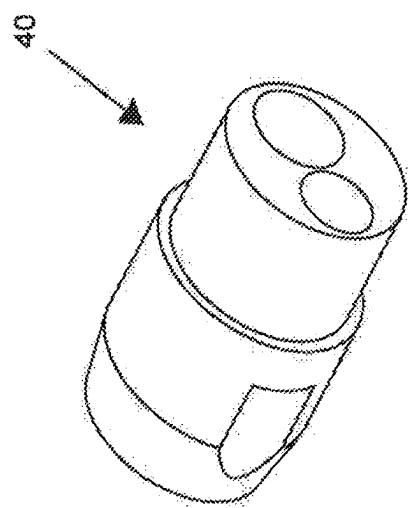


Figure 19

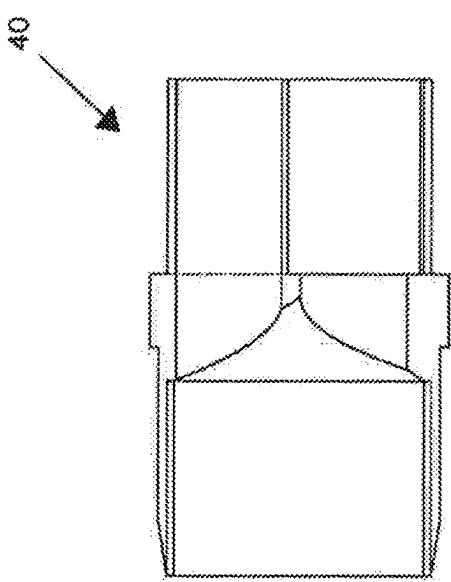


Figure 20

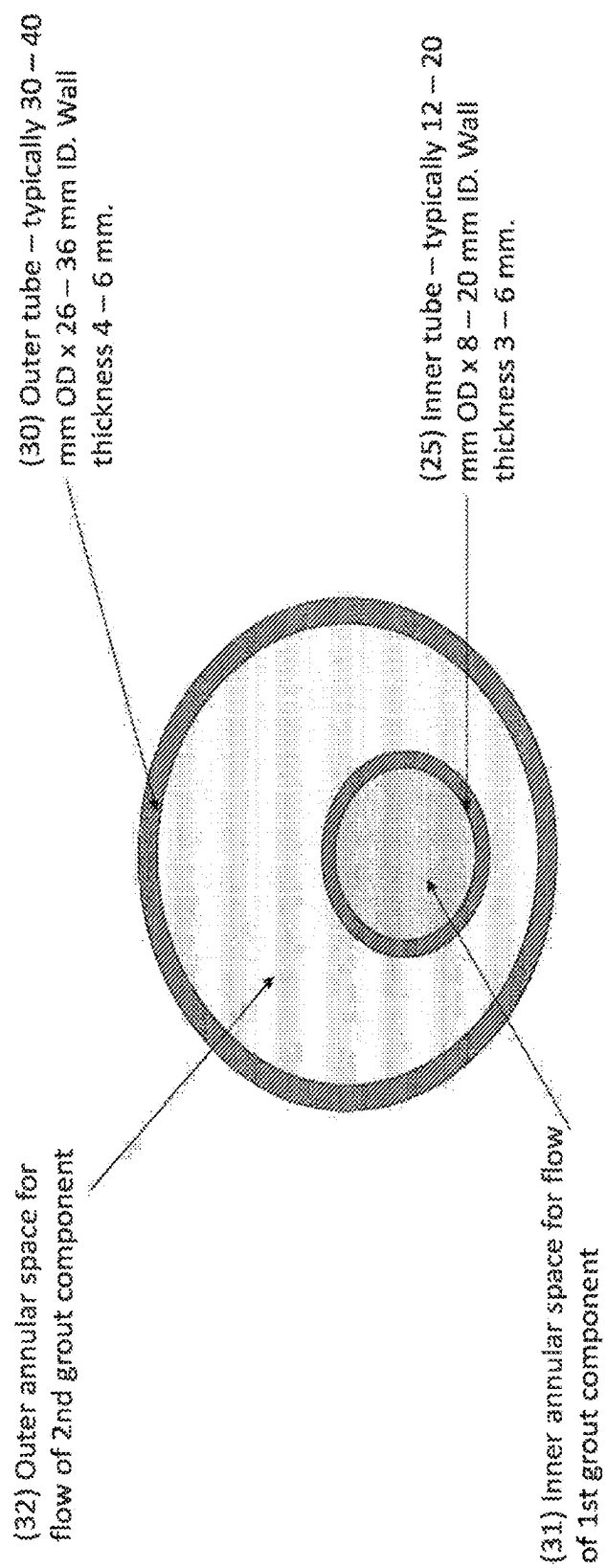


Figure 21

## A METHOD OF INSTALLING A CABLE BOLT, A SYSTEM AND COMPONENTS OF SAME

### TECHNICAL FIELD

[0001] This disclosure relates generally to a method of installing a grouted cable bolt, a system for providing same and components which may form part of the method and/or system.

### BACKGROUND ART

[0002] During the construction of underground excavations such as mines, tunnels and storage caverns, it is normal practice to stabilise the rock surrounding the excavation by means of various forms of rock bolts and anchors that are installed in holes drilled in the rock (referred to as boreholes). The rock bolts consist of rigid elongate elements, such as steel bolts, and the anchors consist of flexible elongate elements, such as steel cables and which are also referred to as cable bolts. The rock bolts or anchors may be coupled to the rock by mechanical friction or by embedment in grout that fills the annulus or gap surrounding the elongate element and the inner surface of the borehole. The grout may be cementitious (that is, principally made from inorganic materials) or resin (falling within the scope of organic chemistry).

[0003] Generally, rock bolts are cheaper and easier and faster to install than cable bolts. They are generally the preferred type. However, if it is necessary to install long anchors, then flexible cable bolts are preferentially used. "Long" in this context means longer than the maximum length that can be conveniently manufactured, transported or installed, given the space constraints at each stage. When very high load-bearing capacity is required, cable bolts may also be preferred as they require no machining during manufacture, therefore higher-strength steel may be used.

[0004] Rock support (whether using rigid or flexible elements) is categorised as "primary" or "secondary". "Primary" support is installed synchronously with the rock excavation process and is intended to stabilise the rock during the same shift as it is installed, for the protection of the personnel engaged in the rock excavation activities. It may also provide long-period support. Rigid support elements (rock bolts) are most commonly used for primary support.

[0005] "Secondary" support is installed at a later time, to provide additional stabilisation of zones identified by rock engineers as requiring extra support. Typical zones requiring installation of secondary support are large-span excavations, long-life excavations and excavations in weak or disturbed rock. As secondary support elements must frequently extend deeper into the rock, to provide the additional stabilisation, and/or must have higher load-bearing capacity, cable bolts are more commonly used. One of the functions of the cable bolt is to provide an axial restraining force if any part of the rock mass traversed by the cable tends to separate, such as by sagging under its own weight.

[0006] The efficacy of the cable bolt in developing this restraining force is measured by the maximum tensile load ("pull-out load") that can be held by a length of cable embedded in grout. This maximum load is limited by the strength of the cable itself. The length of grout embedment that develops a pull-out load equal to the strength of the

cable is called the "critical bond length". The critical bond length for a cable of particular strength is determined by the strength of the embedding grout and by the form of the cylindrical surface of the cable. It is common practice to distort the wires of the cable at selected positions along its length into bulges, called "bulbs" or "birdcages". These increase the pull-out load of a length of cable including at least one bulge embedded in grout.

[0007] The setting performance requirements of the grouts used for secondary support are undemanding. The setting speeds of the grouts are in fact preferably slow, as slow speeds allow more time for preparation and pumping of the grout into the holes before the grout begins to set. After setting, slow strength development is acceptable as the secondary support anchors do not have to provide short-term rock support. The grouts are generally pumped into the pre-drilled holes in the rock through small-diameter "grout tubes" that are attached to the cable anchors themselves, transporting the grout to a distal position in the hole. The open collars of the holes are commonly plugged, preventing the grout from running out. The grouts generally have low viscosity and minimal anti-slump properties.

[0008] As installation of cable bolts is more difficult and slower than installation of rock bolts, the installation of cable bolts is typically part of the secondary support process where they are installed asynchronously from the rock excavation activities. It is generally not necessary to install secondary support everywhere in an underground complex, such as a mine, only in identified zones.

[0009] It will be appreciated that installation of both primary and secondary support is expensive, time consuming and militates against high-efficiency, concentrated mining operations ("mining" referring to any activity that produces underground excavations) as resources must be spread between the primary and secondary support activities that may be a considerable distance apart.

[0010] The present invention seeks to provide an improved method, apparatus and/or system of installing a cable bolt.

### SUMMARY OF THE DISCLOSURE

[0011] In a first aspect embodiments are disclosed of a method of installing a cable bolt within a borehole including an open end and an inner surface leading away from the open end towards a blind end, the method including the following steps:

[0012] (a) delivering a hardenable composition in liquid form at or near the blind end of the borehole and substantially filling the borehole via a delivery assembly with the hardenable composition from the blind end towards the open end of the borehole wherein the hardenable composition is formed from two components, a first component and a second component which when mixed form the hardenable composition in liquid form, the delivery assembly including a first line for delivering the first component and a second line for delivering the second component the first line being located within the second line;

[0013] (b) introducing an anchor end of the cable bolt to the open end of the borehole and pushing the anchor end of the cable bolt through the hardenable composition in liquid form until the anchor end of the cable bolt is proximate to the blind end of the borehole; and,

[0014] (c) allowing the hardenable composition to harden.

[0015] In certain embodiments, the delivery assembly is withdrawn from the borehole at the same, or similar rate as the rate at which the borehole is filled with the hardenable composition.

[0016] In certain embodiments, the first component and the second component are mixed in a mixing chamber in the delivery assembly prior to delivering the hardenable composition in liquid form.

[0017] In certain embodiments, the delivery assembly includes a delivery nozzle at a distal end for delivering the hardenable composition to the borehole.

[0018] In certain embodiments, the mixing chamber is located immediately before the delivery nozzle on the delivery assembly. In one form, the mixing chamber includes a static mixer located therein for assisting with the mixing of the first component and the second component. In a further form, the static mixer is a helical static mixer.

[0019] In certain embodiments, the mixing chamber and the delivery nozzle are located within the borehole during step (a). In one form, the mixing chamber and the delivery nozzle are proximate the blind end of the borehole at the commencement of step (a).

[0020] In certain embodiments, the first line is located within and substantially coaxial with the second line.

[0021] In certain embodiments, the delivery assembly further includes an adaptor with a first port and a second port located at one end, the first port for connecting to a source line of the first component to the adaptor and the second port for connecting a source line of the second component to the adaptor, the adaptor configured to deliver the first component to the first line via the first port, and to deliver the second component to the second line via the second port.

[0022] In certain embodiments, the hardenable composition hardens at least three minutes after step (a) and preferably at least five minutes after step (a).

[0023] In certain embodiment, the blind end is orientated at a higher level than the open end and the hardenable composition includes sufficient anti slump properties whereby the hardenable composition is retained within the borehole after step (a).

[0024] In certain embodiments, the hardenable composition is selected from a two component resin consisting of a resin and a hardener, or a two component cementitious grout.

[0025] In certain embodiments, the method further includes a flushing step after step (a) wherein the first component and second component remaining in the mixing chamber and/or the delivery nozzle of the delivery assembly are flushed therefrom. In one form, the flushing step includes delivering either the first component or the second component to the mixing chamber and delivery nozzle of the delivery assembly.

[0026] In certain embodiments, the borehole has a cross sectional diameter of less than 45 mm.

[0027] In certain embodiments, the delivery assembly has a cross sectional diameter of less than 40 mm. In a further form, the delivery assembly has a cross sectional diameter of less than 35 mm.

[0028] In a further aspect, embodiments are disclosed of a delivery assembly for introducing a hardenable composition in liquid form to a borehole, the delivery assembly including:

[0029] a delivery nozzle at a distal end of the delivery assembly;

[0030] a mixing chamber in fluid communication with the delivery nozzle;

[0031] a first line for conveying a first component to the mixing chamber; and,

[0032] a second line for conveying a second component to the mixing chamber, the first line being located within the second line, and wherein the hardenable composition in liquid form is formed from mixing the first component with the second component.

[0033] In certain embodiments, the first line is located within and substantially coaxial with the second line.

[0034] In certain embodiments, the mixing chamber includes a static mixer located therein for assisting with the mixing of the first component and the second component. In one form, the static mixer is a helical static mixer.

[0035] In certain embodiments, the delivery assembly further includes an adaptor with a first port and a second port located at one end, the first port for connecting to a source line of the first component to the adaptor and the second port for connecting a source line of the second component to the adaptor, the adaptor configured to deliver the first component to the first line via the first port, and to deliver the second component to the second line via the second port.

[0036] In certain embodiments, the delivery assembly further includes a connector in fluid communication with the first line and the second line, the connector configured to deliver the first component and the second component to the mixing chamber.

[0037] In a further aspect, embodiments are disclosed of a system for installing a cable bolt within a borehole including an open end and an inner surface leading away from the open end towards a blind end, the system including the following:

[0038] (i) a delivery assembly for introducing a hardenable composition in liquid form to a borehole, the delivery assembly including a delivery nozzle at a distal end of the delivery assembly; a mixing chamber in fluid communication with the delivery nozzle; a first line for conveying a first component to the mixing chamber; a second line for conveying a second component to the mixing chamber the first line being located within the second line, and wherein the hardenable composition in liquid form is formed from mixing the first component with the second component including a liner open end and a liner blind end;

[0039] (ii) a source of the first component for delivery to the first line; and

[0040] (iii) a source of the second component for delivery to the second line.

[0041] A system according to claim 24 wherein the wherein the first line of the delivery assembly is located within and coaxial with the second line of the delivery assembly.

[0042] Other aspects, features, and advantages will become apparent from the following detailed description when taken in conjunction with the accompanying drawings, which are a part of this disclosure and which illustrate, by way of example, principles of inventions disclosed.

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0043] Notwithstanding any other forms which may fall within the scope of the method and apparatus as set forth in

the Summary, specific embodiments of the method and apparatus will now be described, by way of example, and with reference to the accompanying drawings in which:

**[0044]** FIGS. 1 to 6 are schematic diagrams depicting various steps and/or apparatus in accordance with embodiments of the present disclosure;

**[0045]** FIG. 7 is an elevation view of a delivery assembly in accordance with an embodiment of the present disclosure;

**[0046]** FIG. 8 is a perspective view of the delivery assembly of FIG. 7;

**[0047]** FIG. 9 is an exploded view of part of the delivery assembly in accordance with an embodiment of the present disclosure;

**[0048]** FIG. 10 is an assembled perspective view of the part of the delivery assembly of FIG. 9;

**[0049]** FIG. 11 is a perspective view of the mixing chamber and delivery nozzle of the delivery assembly in accordance with an embodiment of the present disclosure;

**[0050]** FIG. 12 is a cross sectional elevation view of the mixing chamber and delivery nozzle of FIG. 11;

**[0051]** FIG. 13 is section A-A of the mixing chamber and delivery nozzle shown in FIG. 12;

**[0052]** FIG. 14 is a perspective view of a connector of the delivery assembly in accordance with an embodiment of the present disclosure;

**[0053]** FIG. 15 is a cross section elevation view of the connector of FIG. 14;

**[0054]** FIG. 16 is section B-B of the connector shown in FIG. 15;

**[0055]** FIG. 17 is a perspective view of an adapter of the delivery assembly in accordance with an embodiment of the present disclosure;

**[0056]** FIG. 18 is a cross section elevation of the adapter of FIG. 17;

**[0057]** FIG. 19 is section B-B of the adapter shown in FIG. 18;

**[0058]** FIG. 20 is a cross section side elevation of the adapter shown in FIG. 17; and,

**[0059]** FIG. 21 is a cross section view of the first and second line in accordance with an embodiment of the present disclosure.

#### DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS

**[0060]** The present disclosure provides a method of installing a cable bolt within a borehole which enables the method to be carried out during the process of providing primary support. That is the method as herein described may be undertaken at the same time or shortly after the rock excavation process.

**[0061]** The method involves installing a cable bolt in a borehole which includes a circular open end, an inner surface leading away from the open end towards a blind end. Typically, the borehole would be drilled into a rock substrate with the orientation of the holes being from vertical-up to about 30° off the vertical although sometimes the borehole orientation is near to horizontal. The length of the borehole may be up to 20 meters in length but typically borehole lengths are 2 to 6 metres. Typical diameters of the borehole may be from 33 to about 42 mm in length and the cross section is substantially cylindrical providing an inner surface including cylindrical side walls which run the length of the borehole to the blind end. Drilling the borehole may be undertaken by conventional technology and methods.

**[0062]** The method has herein described includes the step of: (a) delivering a hardenable composition in liquid form at or near the blind end of the borehole and substantially filling the borehole with the hardenable composition from the blind end towards the open end of the borehole. This step may be undertaken via a delivery assembly in the absence of the cable bolt within the borehole. Otherwise, stated, the borehole is filled with hardenable composition without the cable bolt located within the borehole.

**[0063]** Once step (a) is completed and the borehole is filled with the hardenable composition, the method moves to step (b) where an anchor end of the cable bolt is introduced to the open end of the borehole. During this step, the cable bolt may be pushed through the hardenable composition which is still in liquid form contained within the borehole until the anchor end of the cable bolt is proximate to the blind end of the borehole. Once the anchor end of the cable bolt is in place proximate to the blind end of the borehole, the method proceeds to step (c) where the hardenable composition is allowed to harden whereby the cable bolt is secured in place.

**[0064]** During step (a) the hardenable composition in liquid form may be delivered at or near the blind end of the borehole via a delivery assembly. The delivery assembly may be withdrawn from the borehole during step (a) at the same, or similar rate as the rate at which the borehole is filled with the hardenable composition. This provides that the hardenable composition does not surround the delivery assembly as the borehole is filled with the hardenable composition.

**[0065]** In certain embodiments, the hardenable composition hardens after at least three minutes after step (a) is completed. In a preferred form, the hardenable composition hardens after at least five minutes after step (a) is completed. In a further preferred form, the hardenable composition hardens within 20 minutes after step (a) is completed. In yet a further preferred form, the hardenable composition hardens within 15 minutes after step (a) is completed.

**[0066]** The hardenable composition may be cementitious or resin and it may be a single component grout or have two or multiple components that react together to initiate hardening.

**[0067]** As herein described, the hardenable composition will generally comprise a mixture of two or more classes of substances, that is at least a first component and a second component, which solidify (harden, or cure) upon mixing. It is preferred that the first component and the second component are also stable in liquid form and are able to be pumped, i.e. pumpable.

**[0068]** The mixing step of the first component and the second component will typically occur during step (a) of the method as herein described such that the hardenable composition is in liquid form for a sufficient time to allow the hardenable composition to harden (cure) during step (c) once the cable bolt is in place within the borehole and surrounded by the hardenable composition. In a preferred form, at least a first component and a second component of the hardenable composition are mixed in a mixing chamber in the delivery assembly prior to delivering the hardenable composition in liquid form.

**[0069]** The hardenable composition may be selected from any one of the following: silicate resins, polyurethane resins, polyester resins, epoxy resins, urea formaldehyde resins, polyurea resins, melamine formaldehyde resins, vinyl ester



resins, furan resins, acrylates, methacrylates, thermosetting resins and/or cement based materials and mortars.

**[0070]** In certain embodiments, additional components such as fillers may be included in the hardenable composition.

**[0071]** It is preferred that the hardening time of the hardenable composition must be delayed for at least 3 minutes, and preferably at least 5 minutes from the time of completion of step (a). This enables enough time for installation of the cable bolt by thrusting through the grout emplaced in the boreholes.

**[0072]** It is preferable that the hardenable composition builds strength rapidly after it has hardened, so that the hardenable composition may form a critical bond length of less than 1 metre within 2 hours, and preferably within 1 hour of completing step (a).

**[0073]** It is preferable that the hardenable composition in liquid form and in particular the first component and the second component have low viscosity of less than 10,000 mPa·s (centipoise) and more preferably, less than 1000 mPa·s, and yet more preferably less than 500 mPa·s. The low viscosity provides benefits as this reduces the pressures needing to be developed to pump the first and second components by their associated pumps. Lower pumping pressures also allows the use of thinner-wall conduits, increasing cross-sectional area available for flow, especially in the confined cross-section of the borehole. Flow of low-viscosity fluids more readily transitions from laminar to turbulent. Turbulent flow enhances mixing, potentially supplementing the effect of the mixing chamber to better form the hardenable composition from mixing the first and second components.

**[0074]** It is also preferable that the hardenable composition rapidly begins “thickening” (i.e. rise in viscosity) where the hardenable composition develops strong “anti-slump” properties upon the mixing of the first component and the second component, so that the grout emplaced in the borehole does not flow out of the hole under its own weight. In a preferred form, the hardenable composition begins thickening within 20 seconds upon mixing of the first component and the second component. In a more preferred form, the hardenable composition begins thickening within 10 seconds upon mixing of the first component and the second component, and in a further preferred time within 5 seconds upon mixing of the first component and the second component.

**[0075]** The “thickening” of the hardenable composition occurs once the first component and the second component are mixed during step (a). The “thickening” of the hardenable composition occurs prior to the “hardening” of the hardenable composition.

**[0076]** More scientifically, the hardenable composition when thickening is preferably a “Bingham Liquid”. “Bingham liquids” require a minimum shear stress before they yield or start to flow. In this case the hardenable composition has a shear stress threshold higher than the stress stresses set up by the weight of the hardenable composition from gravity. Whilst the Bingham characteristic remains desirable, the threshold shear stress should not be so high that the anchor end of the cable bolt cannot be pushed through the hardenable composition when located within a borehole. However, it should be high enough that the cable is retained in the hardenable composition and does not slide out under its own weight.

**[0077]** It is preferred that the rheological properties of the hardenable composition are independent of the hardening of the hardenable composition, which preferably begins after approximately 3 minutes after step (a) and more preferably after 5 minutes after step (a).

**[0078]** It is preferable the hardenable composition has sufficient longevity once cured which provides that the properties of the hardened composition last for at least as long as the design service life of the cable bolt support element. In addition, it is preferable the hardenable composition has a pH that is safe to use, includes corrosion-inhibiting properties, slight expansivity as well as low cost.

**[0079]** In another embodiment, the present disclosure provides a delivery assembly for introducing a hardenable composition in liquid form to a borehole, the delivery assembly including a delivery nozzle at a distal end of the delivery assembly; a mixing chamber in fluid communication and located adjacent the delivery nozzle along the length of the delivery assembly; a first line or conduit for conveying a first component in liquid form of the hardenable composition to the mixing chamber; and, a second line for conveying a second component of the hardenable composition in liquid form to the mixing chamber. The hardenable composition in an initial liquid form may then be prepared within the mixing chamber by mixing the first component with the second component which is then delivered via the delivery nozzle.

**[0080]** The first line may be located within the second line. This provides that the first and second line are contained within a single outside surface. Otherwise stated, the first line is located within the bore of the second line. The first line may be located entirely within the bore of the second line whereby the first line and the second line are in the form of elongate conduits where an outside cylindrical surface of the first line is entirely contained within and separate to an inner circumferential surface of the second line. In an alternative form, the first line may be located within the bore of the second line where at least a part of the inner circumferential surface of the first line is formed from the second line.

**[0081]** The first line may be located substantially in line with the axis of the second line such that the first line may be within and substantially coaxial with the second line. Alternatively, the first line may be located within the second line and where the first and second line do not share the same axis.

**[0082]** By providing a structure where the first line is located within the second line the first and second line have a more streamlined outer surface (i.e. the surface of the second line) which assists in inserting and withdrawing the delivery assembly within the borehole in line with the method as herein described. In this embodiment, the first line may be a cylindrical elongate conduit with an outer surface. The first line may be contained within the second line which is also in the form of a cylindrical elongate conduit. This means that the outside surface of the first line is contained within the interior surface of the second line. This structure also provides that the second component flowing through the second line is in contact with the outside surface of the first line, but not in contact with the first component flowing through the first line.

**[0083]** An advantage of the first line being located within the second line is that this structure makes better use of the available cross-sectional area within the borehole, allowing

more of the cross-sectional area to be utilised for flow of the liquid components through the first and second lines. This reduces the pumping pressures required and/or enables higher flow rates of the first and second component. In addition, the structure provides that the pressure difference across the wall of the first line is less than the pressure difference experienced if the first line was external from the second line. This is because the outer surface of the first tube is subject to the pressure of the fluid flowing in the surrounding second line, which may be close to or substantially similar to the pressure of the fluid in the first line. This allows that the wall thickness of the first line may be reduced.

**[0084]** A further advantage that is provided when the first line is located within the second line is that this arrangement reduces the outside circumference of the first and second line which provides that the delivery assembly has a reduced cross sectional area. By having a reduced cross-sectional area, it is possible to insert the delivery assembly in boreholes with diameters less than 45 mm in diameter and less than 33 mm in diameter.

**[0085]** An example of the first line **25** located within the second line **30** is depicted in FIG. **21** showing a cross section of the structure. The second component **32** is shown flowing through the internal bore of the second line **30** and surrounding the outside surface of the first line **25**. The first component **31** is shown flowing through the internal bore of the first line **21**.

**[0086]** The mixing chamber of the delivery assembly may also include a static mixer located therein for assisting with the mixing of the first component and the second component. The static mixer may be in the form of a helical static mixer.

**[0087]** The delivery assembly may also include an adaptor with a first port and a second port located at one end, the first port may be able to connect to a source line of the first component which may in the form of a fluid conduit leading to a tank or reservoir containing the first component. The second port of the adaptor may be able to connect to a source line of the second component which may be in the form of a fluid conduit leading to a tank or reservoir containing the second component. A pump may be attached to the source line of the first component and another pump may be attached to the source line of the second component to assist in delivering the fluids to the first and second ports of the adaptor. The adaptor is structured to then deliver the first component to the first line via the first port, and to deliver the second component to the second line via the second port.

**[0088]** The delivery assembly may also include a connector which is in fluid communication with the first line and the second line and located immediately before the mixing chamber along the length of the delivery assembly. The connector may include an inlet at one end of the connector which is in fluid communication with the first line and the second line and a deliver port at the other end of the connector which is in fluid communication with the mixing chamber. The connector functions to deliver the first component and the second component from the first line and the second line to the mixing chamber.

**[0089]** In another embodiment, the present disclosure provides a system for installing a cable bolt, the system including the following: a delivery assembly as herein described for introducing a hardenable composition in liquid form to

a borehole, a source of the first component for delivery to the first line; and a source of the second component for delivery to the second line.

**[0090]** The system may also include a mechanised “rig” for drilling of holes in rock, the pumping of hardenable composition into the holes and the thrusting of cable anchors through the already-pumped hardenable composition, to span substantially the entire length of the borehole. The rig (and similar rigs) develops the force to thrust the cable into the boreholes by passing the cable between two powered rollers that rotate in unison, engaging frictionally with the cable to advance it. To develop high thrust, the rollers exert a strong compressive force on the cable as it passes between them.

**[0091]** The method and system as described herein will now be described with reference to an example embodiment depicted in FIGS. **1** to **6** which depict various sequential stages, and the apparatus used in those stages, during the installation of a cable bolt.

**[0092]** Referring to FIG. **1** a borehole **100** is drilled in a vertical orientation into rock strata **120**. The borehole **100** includes an opening **105** at one end and a blind end **110** at the other end with side walls **115** which are cylindrical in form leading from the opening **105** to the blind end **110**. Typically, the orientation of the borehole is vertical leading from the opening **105** to the blind end **110**. The borehole may also be at any angle such as from about 30° off the vertical and sometimes extending to near horizontal. Typical borehole lengths may be 2.5-6 metres from the opening to the blind end and borehole diameters may be from 33-42 mm. Drilling the borehole may be accomplished by conventional technology and methods such as using a drill rig and associated drill string in the form of multiple drill rods. After completion of drilling, the drilling rods are withdrawn and removed from the borehole. The drilling rig may remain in position after drilling the borehole, so that it is aligned with the hole for execution of the subsequent steps of the method.

**[0093]** Referring to FIG. **1**, the drilling rig introduces the delivery assembly **10** for delivering the hardenable composition. The delivery assembly **10** includes a delivery nozzle **15**, mixing chamber **20** and the first and second lines **25**, **30** which in the embodiment shown, has the outside appearance of a single line where the first line **25** is located and contained within the second line **30**. The cross sectional diameter of the delivery assembly is less than the diameter of the borehole, i.e. the cross sectional diameter of the delivery assembly is less than 40 mm, and in this embodiment is less than 35 mm in diameter. The delivery nozzle **15** is positioned at or immediately before the blind end **110** of the borehole **100**. At this stage in the method, the cable bolt has not been introduced to the borehole.

**[0094]** Referring to FIG. **3**, the next stage in the method includes the delivery of the hardenable composition **130** to the borehole **100**. The hardenable composition **130** is formed by mixing a first component with a second component delivered by the first and second lines **25**, **30** to the mixing chamber **20**. The hardenable composition **130** is then injected into the borehole **100** from the delivery nozzle **15** starting from the blind end **110**. The first component and the second component flow from their respective pumps through separate lines and then through the first and second lines **25**, **30** to the mixing chamber **20**. The separate lines may include control and measurement components such as valves,

flowmeters and pressure sensors. In view of their large size, these devices remain external to the delivery assembly 10 which enters the length of the borehole 100. FIG. 3 also shows that the delivery assembly 10 including the delivery nozzle 15 is withdrawn from the blind end 110 of the borehole 100 towards the opening 105. It is preferable to match the speed of withdrawal to the rate of extrusion of the hardenable composition, so that the void left by withdrawal of the delivery nozzle is filled with hardenable composition. Typical withdrawal rates are of 3-10 m per minute.

[0095] Referring to FIG. 4, after a pre-determined volume of hardenable composition has been extruded, further delivery of the hardenable composition is stopped. Withdrawal of the delivery assembly 10 continues until it is completely withdrawn from the opening 105 of the borehole 100.

[0096] At this stage and in reference to FIG. 5, the mixing chamber 20 of the delivery assembly 10 contains hardenable composition that is at least partially mixed and formed from the first component and the second component. If left in the mixing chamber and delivery nozzle, the hardenable composition will harden and cause blockages. It is therefore preferred to undertake a flushing step of the delivery assembly 10. One of the first or second components may be used as the flushing medium. Alternative flushing media, such as water or oil may be used. However, use of a distinct flushing medium requires that the hardenable composition piping system include an additional line for the flushing media and appropriate valving and pumping systems. Typically, 80-100 millilitres of flushing medium are required to clear the (partially) mixed hardenable composition out of the mixing chamber 20 and the delivery nozzle 15.

[0097] Referring to FIG. 6, the drilling rig then introduces the cable bolt 140 into through the opening 105 of the borehole 100 and pushes the anchor end 145 of the cable bolt 140 through the hardenable composition 130 located in the borehole 100 until the anchor end 145 reaches the blind end 110 of the borehole 100. This operation may take place after or simultaneously with flushing of the delivery assembly as shown in FIG. 5. In any event, the insertion of the cable bolt 140 must be completed before the hardenable composition in the borehole has set. This is preferentially 4-10 minutes from commencement of pumping of the hardenable composition as depicted in FIG. 3.

[0098] Referring to FIGS. 7 and 8 there is shown an embodiment of the delivery assembly 10 in accordance with an embodiment. The delivery assembly 10 includes a delivery nozzle 10 at one end that is in fluid communication with a mixing chamber 20. The mixing chamber 20 is in fluid communication with the first line and second line 25, 30 which in this embodiment are in the form of a coaxial line. A connector 60 provides delivery of the first component and second component from the first line and second line 25, 30 to the mixing chamber 20 where the first component and second components are mixed before forming the hardenable composition which is then delivered via the delivery nozzle 15. The first line and second line 25, 30 are in fluid communication and joined to an adapter 40 which includes a first port 50 and a second port 55 on one end of the adapter 40. The first port 50 and second port 55 provide a connection point to deliver of the first and second components respectively from a source of the first component and a source of the second component. The adapter 40 thereby delivers the first component and the second component to the first line and second line 25, 30 respectively.

[0099] The mixing chamber 20 and delivery nozzle 15 are shown in more detail in FIGS. 9, 10, 11 and 20. The first component and second component are delivered to the mixing chamber 20 via the connector 60 which is in fluid communication with the first line and second line 25, 30. The mixing may be facilitated in the mixing chamber 20 by a static mixer 35 (not shown) but which may be located in the cavity best seen in FIG. 13. The static mixer 35 may be in the form of a helical static mixer.

[0100] Referring to FIGS. 14, 15 and 16 there is shown further detail of the connector 60 of the delivery assembly 10. The connector 60 includes an inlet port 65 for receiving the flow of fluid from the first and second line 25, 30 and a delivery port 70 for delivering the flow of the first and second components to the mixing chamber 20. To enable the delivery of the first component from the first line which is within and coaxial with the second line, and the second component from the second line, the connector includes two openings 71, 72 in the delivery port 70. The first opening 71 generally aligns with the delivery of the first component from the first line 25 located within and coaxial with the second line 30. The first opening 71 is in the form of a generally circular opening in a central region of the delivery port 70. The second opening 72 aligns with at least part of the delivery of the second component from the second line 30. The second opening 72 is located in an outer circumferential region and extends around part of the area near the circumference of the delivery port.

[0101] The adaptor 40 is shown in further detail in FIGS. 17, 18 19 and 20.

[0102] In the foregoing description of preferred embodiments, specific terminology has been resorted to for the sake of clarity. However, the invention is not intended to be limited to the specific terms so selected, and it is to be understood that each specific term includes all technical equivalents which operate in a similar manner to accomplish a similar technical purpose. Terms such as “top” and “bottom”, “front” and “rear”, “inner” and “outer”, “above”, “below”, “upper” and “lower” and the like are used as words of convenience to provide reference points and are not to be construed as limiting terms.

[0103] The reference in this specification to any prior publication (or information derived from it), or to any matter which is known, is not, and should not be taken as, an acknowledgement or admission or any form of suggestion that prior publication (or information derived from it) or known matter forms part of the common general knowledge in the field of endeavour to which this specification relates.

[0104] In this specification, the word “comprising” is to be understood in its “open” sense, that is, in the sense of “including”, and thus not limited to its “closed” sense, that is the sense of “consisting only of”. A corresponding meaning is to be attributed to the corresponding words “comprise”, “comprises” and “comprises” where they appear.

[0105] In addition, the foregoing describes only some embodiments of the invention(s), and alterations, modifications, additions and/or changes can be made thereto without departing from the scope and spirit of the disclosed embodiments, the embodiments being illustrative and not restrictive.

[0106] Furthermore, invention(s) have been described in connection with what are presently considered to be the most practical and preferred embodiments, it is to be understood that the invention is not to be limited to the disclosed

embodiments, but on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the invention(s). Also, the various embodiments described above may be implemented in conjunction with other embodiments, e.g., aspects of one embodiment may be combined with aspects of another embodiment to realize yet other embodiments. Further, each independent feature or component of any given assembly may constitute an additional embodiment.

Table of Parts

|                        |     |
|------------------------|-----|
| Delivery assembly      | 10  |
| Delivery nozzle        | 15  |
| Mixing chamber         | 20  |
| First line             | 25  |
| Second line            | 30  |
| Static Mixer           | 35  |
| First component        | 31  |
| Second component       | 32  |
| Adapter                | 40  |
| First port             | 50  |
| Second port            | 55  |
| Connector              | 60  |
| Inlet port             | 65  |
| Delivery port          | 70  |
| First opening          | 71  |
| Second opening         | 72  |
| Borehole               | 100 |
| Opening                | 105 |
| Blind end              | 110 |
| Side walls             | 115 |
| Rock strata            | 120 |
| Hardenable composition | 130 |
| Cable bolt             | 140 |
| Anchor end             | 145 |

1. A method of installing a cable bolt within a borehole including an open end and an inner surface leading away from the open end towards a blind end, the method including the following steps:

- (a) delivering a hardenable composition in liquid form at or near the blind end of the borehole via a delivery assembly and substantially filling the borehole with the hardenable composition from the blind end towards the open end of the borehole wherein the hardenable composition is formed from two components, a first component and a second component which when mixed form the hardenable composition in liquid form, the delivery assembly including a first line for delivering the first component and a second line for delivering the second component the first line being located within the second line;
- (b) introducing an anchor end of the cable bolt to the open end of the borehole and pushing the anchor end of the cable bolt through the hardenable composition in liquid form until the anchor end of the cable bolt is proximate to the blind end of the borehole; and,
- (c) allowing the hardenable composition to harden.

2. The method according to claim 1 wherein the delivery assembly is withdrawn from the borehole at one of a same or similar rate as the rate at which the borehole is filled with the hardenable composition.

3. The method according to claim 1 wherein the first component and the second component are mixed in a mixing chamber in the delivery assembly prior to delivering the hardenable composition in the liquid form;

the delivery assembly includes a delivery nozzle at a distal end for delivering the hardenable composition to the borehole; and

the first line and the second line deliver the first component and the second component to the mixing chamber which is in fluid communication with the delivery nozzle, with the mixing chamber being located immediately before the delivery nozzle on the delivery assembly.

4. (canceled)

5. (canceled)

6. The method according to claim 3 wherein the mixing chamber is located immediately before the delivery nozzle on the delivery assembly, with the mixing chamber includes a static mixer located therein for assisting with the mixing of the first component and the second component, with the static mixer is a helical static mixer.

7. (canceled)

8. (canceled)

9. The method according to claim 3 wherein the mixing chamber and the delivery nozzle are one or more of located within the borehole during step (a) and proximate the blind end of the borehole at a commencement of step (a).

10. (canceled)

11. The method according to claim 1 wherein the first line is located within and substantially coaxial with the second line.

12. The method according to claim 11 wherein the delivery assembly further includes an adaptor with a first port and a second port located at one end, the first port for connecting to a source line of the first component to the adaptor and the second port for connecting a source line of the second component to the adaptor, the adaptor configured to deliver the first component to the first line via the first port, and to deliver the second component to the second line via the second port.

13. The method according to claim 1 wherein the hardenable composition hardens at least three minutes after step (a) and preferably at least five minutes after step (a).

14. The method according to claim 1 wherein the blind end is orientated at a higher level than the open end and wherein the hardenable composition includes sufficient anti slump properties whereby the hardenable composition is retained within the borehole after step (a).

15. The method according to claim 1 wherein the hardenable composition is selected from a two component resin consisting of a resin and a hardener, or a two component cementitious grout.

16. The method according to claim 3 wherein the method further includes a flushing step after step (a) wherein at least one of the first component and second component remaining in at least one of the mixing chamber and the delivery nozzle of the delivery assembly are flushed therefrom.

17. (canceled)

18. The method according to claim 1 wherein the borehole has a cross sectional diameter of less than 45 mm.

19. (canceled)

20. (canceled)

21. A delivery assembly for introducing a hardenable composition in liquid form to a borehole, the delivery assembly including:

- a delivery nozzle at a distal end of the delivery assembly;
- a mixing chamber in fluid communication with the delivery nozzle;

a first line for conveying a first component to the mixing chamber; and  
a second line for conveying a second component to the mixing chamber the first line being located within the second line, and wherein the hardenable composition in liquid form is formed from mixing the first component with the second component.

**22.** The delivery assembly according to claim **21** wherein the first line is located within and substantially coaxial with the second line.

**23.** The delivery assembly according to claim **21** wherein the mixing chamber comprises a static mixer located therein for assisting with the mixing of the first component and the second component, with the static mixer being a helical static mixer.

**24.** (canceled)

**25.** The delivery assembly according to claim **21** further including an adaptor with a first port and a second port located at one end, the first port for connecting to a source line of the first component to the adaptor and the second port for connecting a source line of the second component to the adaptor, the adaptor configured to deliver the first component to the first line via the first port, and to deliver the second component to the second line via the second port.

**26.** The delivery assembly according to claim **21** further including a connector in fluid communication with the first line and the second line, the connector configured to deliver the first component and the second component to the mixing chamber.

**27.** The delivery assembly according to claim **21** wherein the delivery assembly has a cross sectional diameter of less than 40 mm.

**28.** (canceled)

**29.** A system for installing a cable bolt within a borehole including an open end and an inner surface leading away from the open end towards a blind end, the system including the following:

- i. a delivery assembly for introducing a hardenable composition in liquid form to a borehole, the delivery assembly including a delivery nozzle at a distal end of the delivery assembly; a mixing chamber in fluid

communication with the delivery nozzle; a first line for conveying a first component to the mixing chamber; a second line for conveying a second component to the mixing chamber the first line being located within the second line, and wherein the hardenable composition in liquid form is formed from mixing the first component with the second component including a liner open end and a liner blind end;

- ii. a source of the first component for delivery to the first line; and

- iii. a source of the second component for delivery to the second line.

**30.** A system according to claim **29** wherein the first line of the delivery assembly is located within and coaxial with the second line of the delivery assembly.

**31.** The system according to claim **29** wherein the mixing chamber of the delivery assembly includes a static mixer located therein for assisting with the mixing of the first component and the second component, with the static mixer being a helical static mixer.

**32.** (canceled)

**33.** The system according to claim **29** wherein the delivery assembly further includes an adaptor with a first port and a second port located at one end, the first port for connecting to the source of the first component to the adaptor and the second port for connecting the source of the second component to the adaptor, the adaptor configured to deliver the first component to the first line via the first port, and to deliver the second component to the second line via the second port.

**34.** The system according to claim **29** wherein the delivery assembly further includes a connector in fluid communication with the first line and the second line, the connector configured to deliver the first component and the second component to the mixing chamber.

**35.** The system according to claim **29** wherein the delivery assembly has a cross sectional diameter of less than 40 mm.

**36.** (canceled)

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